

AJAY PRANAV BALAKRISHNAN

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PROFESSIONAL SUMMARY

Computer Science undergraduate at VIT Vellore who builds and deploys full-stack and AI-powered applications in Python (Django, Flask, FastAPI) and JavaScript (React). Experienced in LLM API integration (Anthropic Claude, Google Gemini), relational database design, OAuth 2.0 authentication, WebSocket dashboards, and applied ML pipelines using transfer learning and vector search. Seeking backend, software development, or full-stack internships.

EDUCATION

Vellore Institute of Technology (VIT), Vellore

2025 – 2029

Bachelor of Technology — Computer Science & Engineering

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

AlgoTrader — AI-Driven Intraday Trading System

github.com/Ajaypranav25/algotrader

Python · FastAPI · React · Vite · WebSockets · Google Gemini API · Angel One SmartAPI · SQLAlchemy

- Built an automated intraday trading system for Indian equities that uses Google Gemini for signal generation and Angel One SmartAPI for market data and order execution, driven by a background trading loop
- Implemented a risk-management layer with per-trade capital caps, a daily-loss halt, trailing stop-loss, a market-hours guard, and a one-click kill switch that squares off all positions; paper-trading mode is on by default
- Developed a React (Vite) dashboard showing live watchlist, positions, trade log, AI analysis, and P&L chart, streamed from the FastAPI backend over WebSockets, with encrypted credential storage

TaskMarket — AI-Evaluated Task Marketplace

github.com/Ajaypranav25/TaskMarket

Python · Django · SQLite · Anthropic Claude API · WhiteNoise · HTML/CSS/JS

- Built a full-stack Django platform where companies post tasks and workers complete them for rewards, with role-aware dashboards, submission review, and a real-time leaderboard
- Integrated the Anthropic Claude API to score every submission from 0–100 with written feedback, replacing manual review with consistent, automated evaluation
- Designed a five-tier level system (Rookie → Master) that gates tasks by average score and submission count, with dynamic payouts: $\text{reward} \times (\text{score}/100) \times \text{level multiplier}$
- Modeled users, levels, tasks, and submissions in a relational schema, with Django admin and a demo-data seeding command

Index — Local Semantic File Search Engine

github.com/Ajaypranav25/ai-file-classifier

Python · FastAPI · PyTorch · OpenCLIP · scikit-learn · LanceDB · Tesseract OCR · Watchdog

- Built a local-first background service that watches folders and runs an extract (OCR / PDF / DOCX) → CLIP embed → classify → store pipeline in real time, with no cloud dependency
- Applied transfer learning by training a scikit-learn LogisticRegression on frozen OpenCLIP (ViT-B-32) embeddings, with a zero-shot CLIP fallback for cold start, instead of training a model from scratch
- Built a hybrid semantic + keyword search API (FastAPI) over an on-disk LanceDB store, with an active-learning loop that retrains the classifier from user corrections in under a second and hot-reloads it without a restart

VIT CabShare — Institutional Ride-Sharing Platform

vitcabshare.vercel.app

Python · Flask · SQLAlchemy · Supabase (PostgreSQL) · Google OAuth 2.0 · Vercel

- Built and deployed a cab-sharing app for VIT students, restricting access with Google OAuth 2.0 (Authlib) and institutional email-domain validation, with session checks on every protected route
- Designed a relational schema (User, Ride, Booking) with validation that prevents double bookings and self-booking, tracks live seat availability, and restores seats automatically on cancellation

Smart Local Screenshot Search

github.com/Ajaypranav25/Smart-Local-Screenshot-Search-Engine

Python · FastAPI · Tesseract OCR · SQLite FTS5 · sentence-transformers · pytest

- Built an offline screenshot search app (OCR, embeddings, categorization) that merges vector and full-text results with Reciprocal Rank Fusion; wrote 19 tests, including an end-to-end watcher → OCR → search test

Additional Projects: **RESTful Online Voting System** (Flask, Jinja2) — vote deduplication, server-side validation, real-time result aggregation; **Task Manager** (Flask, SQLite) — CRUD app with RESTful routing and schema migrations.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, C, C++

Backend & APIs: Django, Django REST Framework (DRF), Flask, FastAPI, SQLAlchemy, RESTful API Design, WebSockets, Pydantic

Frontend: React.js, Vite, HTML5, CSS3

Databases: PostgreSQL (Supabase), SQLite (FTS5), LanceDB, Relational Database Design, ORM

AI & ML: Anthropic Claude API, Google Gemini API, PyTorch, OpenCLIP / Transfer Learning, scikit-learn, Vector Search, Tesseract OCR

Auth & Security: Google OAuth 2.0 (Authlib), JWT, Session Management

Tools & Concepts: Git, GitHub, Vercel, Postman, pytest, Linux/Unix CLI, VS Code, System Design, MVC, SDLC, Agile